When Is Video Enough? Predicting Deformable Object Responses to New Interactions

Anonymous CVPR submission

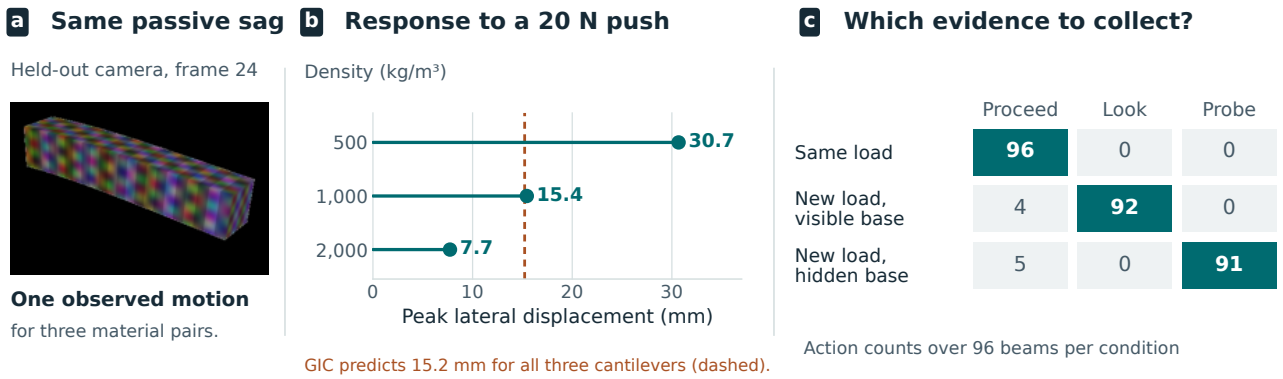


Figure 1. **The next interaction determines what the robot needs to know.** In the left and centre panels, three simulated cantilevers have the same sag, yet move by 7.7–30.7 mm under a 20 N lateral push. GIC fitted to the passive images predicts 15.2 mm in all three. The right panel shows how, in a separate 96-case beam study, the initial tip observation can justify proceeding, observing the support, or probing, depending on the target and the accessible view.

Abstract

A renderable visual–physical world model should predict what happens under new interactions, not merely reproduce the appearance of observed motion. Different material, support and loading explanations can generate the same video yet disagree about the response to a future action. We formulate this as an action-conditioned physical ambiguity problem: retain plausible mechanical hypotheses, propagate their outcomes under the intended intervention, estimate response-error risk, and decide whether to proceed, observe another view or apply a calibrated probe. When the observation map is affine in unknown force amplitudes, we marginalize these forces analytically conditional on spatial material and support hypotheses; with known commands, the posterior is propagated by physical rollouts. We evaluate three prospectively sealed RGB benchmark tasks: an oblique force on deformable cantilevers, an off-centre press on a block with a concealed stiff inclusion, and a freely rebounding striker. After one calibration episode our method yields 1.27, 0.33 and 0.51 mm mean 3D displacement error, respectively, and 0.003 N s mean axial impulse error. The lowest competing errors are 5.26, 1.28 and 3.47 mm, and 0.069 N s. A withheld-camera analysis reveals how rendering quality can obscure physically different predictions. Independent 96-case measurement decisions and spatial material/support/load studies demonstrate the value of acquiring action-specific evidence, while model-mismatch controls expose the sensitivity of uncertainty estimates to their physical forward models.

1. Introduction

A central ambition of visual–physical world models is to reconstruct a scene from visual observations and predict its evolution under novel physical interactions, while preserving the ability to render it from arbitrary viewpoints. This joint ambition connects dynamic neural rendering to physical simulation and action-conditioned prediction. Dynamic Gaussian representations have made temporally coherent 3D reconstruction fast and photorealistic [26, 28]; physically grounded Gaussian and radiance-field systems bring explicit simulation into the representation [9, 15, 27]; and learned world models increasingly generate futures conditioned on manipulations or partial geometric controls [10, 12]. Together, these developments promise interactive digital twins for robotics and embodied reasoning.

However, *reconstructing observed motion and predicting a new physical response are different inverse problems.*

A method may explain the appearance, shape, and motion of the training sequence accurately while remaining wrong about the response to an unobserved force, contact, or support change. This is not necessarily an optimization or simulation defect. Several physically distinct worlds can generate the same observed trajectory, particularly when material stiffness, inertia, applied loads, and boundary compliance are incompletely known. No amount of photometric refinement of an intrinsically ambiguous trajectory can identify the missing mechanical scale. Accordingly, held-view PSNR and perceptual video quality are insufficient proxies for the accuracy of *counterfactual* force–displacement or impulse predictions. Recent physics-oriented benchmarks further motivate evaluating the physical state and response rather than rendered plausibility alone [5, 11].

The unresolved ambiguity is also *action dependent*. In Fig. 1, three simulated cantilevers sag indistinguishably under gravity because their density and stiffness scale together. A known lateral push produces peak deflections from 7.7 to 30.7 mm, but repeating the original passive motion does not separate the worlds. The issue is therefore not whether every material parameter can be recovered uniquely. It is whether the explanations compatible with the available video predict sufficiently similar outcomes for *the action the robot intends to take*. Existing models already support novel interactions in important regimes [6, 9, 23, 25]; what remains insufficiently resolved is a prospective, tolerance-aware account of *when* an individual prediction is supported by its observations.

This viewpoint makes evidence acquisition part of physical world modeling. An additional camera may reveal out-of-plane deformation or a moving support, but it cannot necessarily recover an absolute stiffness scale from the same passive loading. A known-force probe may break that ambiguity, yet can be unnecessary when the intended action is already predictable. We seek to decide whether to *proceed, look, or probe* by comparing how each option is expected to change the error of the specified future response, rather than maximising visual reconstruction quality or parameter information independently of the task.

We propose a task-conditioned inference and decision layer for renderable physical twins. In the spatial identification setting, Gaussian fields parameterize uncertain material variation, support compliance, and load footprints in a finite-element model; Gaussian integration eliminates unknown load amplitudes where the observation map is affine in force. In known-command dynamic settings, calibrated image-derived surface motion weights candidate mechanical models, which are rolled out under a new force or contact. Both settings produce a distribution of possible *responses*, from which we estimate tolerance-exceedance risk and the value of an additional measurement. Our experiments test prospective physical transfer from RGB motion, joint material/support/load inference, and cost-aware evidence selection. Our contributions are (i) an action-specific physical ambiguity and risk formulation, (ii) a joint material/support/load posterior with task-conditioned view-or-probe selection, and (iii) prospective new-force, hidden-inclusion and free-impact predictions with independently assessed rendering and uncertainty.

2. Related Work

Dynamic 3D and 4D rendering. Deformable 3D Gaussians, 4D Gaussian Splatting, and Spacetime Gaussian Feature Splatting reconstruct time-varying scene appearance and support novel-view or novel-time synthesis [16, 26, 28]. These representations are powerful accounts of *where the scene appeared to move*; their standard deformation fields are indexed by time rather than a measured force or new robot command. *Limitation:* rendering an unseen viewpoint or interpolating a timestamp does not establish a physically valid response to an unseen intervention.

Physics-enabled rendering and forward simulation. PhysGaussian unifies 3D Gaussian rendering with material-point dynamics; Spring-Gaus couples Gaussian appearance to spring-mass systems; and PhysDreamer draws on video-generation priors to animate reconstructed assets under interactions [27, 30, 33]. PhysFlow, PhysSplat, GaussianFluent, and the learned Neural Gaussian Force Field extend simulation to richer materials, damage, or more efficient dynamics, while PhysGM predicts a simulation-ready Gaussian asset in a feed-forward pass [8, 14, 17, 18, 31]. Several methods genuinely accept physical controls or synthesize new interactions. *Limitation:* physical plausibility under a configured simulator or generative prior is not, by itself, calibrated evidence that its predicted response to a particular real-world load is correct.

Joint reconstruction and inverse physical identification. PAC-NeRF and GIC fit physical parameters through video, simulation, and rendering; MASIV learns a constitutive model from observed motion [3, 15, 32]. PhysTwin simulates changed grasp trajectories, M-PhyGs estimates spatially varying material for transfer between manipulations, and ReconPhys predicts monocular-video-derived appearance and physical attributes [9, 23, 24]. Most directly, PhysConvex uses deformable convexes as common rendering and continuum-simulation supports, jointly refines physical parameters from multiview videos, quantitatively evaluates eight held-out continuation frames and illustrates modified forces and boundaries [25]. Its published new-force and new-boundary demonstrations should not be dismissed as absent, but neither should they be equated with prospectively calibrated force-response risk across ambiguous physical worlds. *Limitation:* these methods provide capable fitted physical

twins, but their reported evaluations do not directly test whether a particular next response remains non-identifiable under several observationally compatible material, load, and support configurations.

Action-conditioned and generative world models. P3Sim conditions a probabilistic RGB/depth/flow model on partial geometric motion signals derived from specified camera or object transforms; its quantitative benchmarks address novel-view synthesis and image-based 3D editing, whereas its deformable manipulation and collision demonstrations are primarily qualitative [12]. Physical Object Understanding learns a physically controllable probabilistic object model [22]. Dexterous World Models predicts hand-motion-conditioned video; Goal Force explicitly conditions on force-related goals, PhyCo conditions on physical properties, and video–action models condition on robot actions [7, 10, 19, 34]. Thus, *missing action conditioning* is not the gap. *Limitation:* geometric edits, generated futures, and action-progress scores are not evaluations of calibrated millimetre-level deformation or Newton-valued reaction-force error across observationally ambiguous mechanical explanations.

Interactive calibration, uncertainty, and active measurements. Latent-force Gaussian processes and Bayesian experimental design provide ways to infer unmeasured forcing and choose informative observations [1, 2, 20, 21]. KnockGS calibrates two global effective elasticity/density scales from a known Probe-A particle response, freezes its response-feature regression estimate and tests unseen probes of different direction and magnitude [6]. This is a direct and important probe-to-probe prediction precedent, though its object-specific candidate-response library and partial released code do not establish a general end-to-end image-to-simulator pipeline. PhysCoRe infers spatial material from observed RGB-D motion and corrects an MPM solver with a learned velocity residual; it reports real-object dynamics prediction and visualises material-confidence changes as a robot probes objects [29]. The published confidence-guided exploration section does not report a matched-budget quantitative policy comparison that prospectively selects among view, probe and proceed. *Limitation:* both works address important components of interaction-informed prediction, but neither explicitly minimises tolerance-exceedance risk for the *specified* next action over alternative evidence sources and their acquisition costs.

Physical-fidelity datasets and evaluations. PhysGaia evaluates multi-body physical consistency in dynamic novel-view synthesis, PhysInOne provides broad annotated physical phenomena, and Deform360 combines multiview visuotactile evidence for deformable-world modeling [11, 13, 35]. GaussianBench further separates physical failures, unsupported capabilities, and invalid tests under interventions [5]. *Limitation:* broad reconstruction or physics-fidelity evaluations do not themselves answer, prospectively, whether the currently observed object is predictable *for one specified next action*, nor whether looking or probing is the cheaper way to make that prediction reliable.

3. Predicting an Interaction from Partial Evidence

Let D contain the available motion observations and known input commands. A physical explanation z consists of the relevant material, boundary, and loading variables. A forward model produces the visible observations $g(z)$ and the response $q(z, a_\star)$ to a specified future interaction $a_\star$. The target may be a trajectory of surface points or an impulse delivered during contact. The command is available at prediction time. Its realized response is withheld.

When does an ambiguity matter? Suppose two explanations z_1, z_2 produce the same noiseless observations. Every estimator using those observations and the same target command must give them the same point prediction $\hat{q}$. For a response metric d the triangle inequality gives

$$\max_{i=1,2} d\big(q(z_i, a_\star), \hat{q}\big) \;\geq\; \tfrac{1}{2} d\big(q(z_1, a_\star), q(z_2, a_\star)\big). \quad (1)$$

If the two responses differ by more than 2δ, no single estimate can meet tolerance δ in both worlds. In Fig. 1, even a perfect passive fit must miss at least one extreme cantilever by about 11.5 mm in peak lateral deflection. Changing the task to a repeated gravity release collapses this separation. This bound is a worst-case statement. Estimating the probability of a threshold violation additionally requires a distribution over the plausible explanations.

Prediction and a decision to measure. We assign posterior weights to the physical hypotheses using D. Under squared error the point forecast is the posterior mean, $\hat{q} = \mathbb{E}[q(z, a_\star) \mid D]$. For any specified point forecast, its estimated task risk is

$$r(D, a_\star) = \Pr\big[d\big(q(z, a_\star), \hat{q}(D, a_\star)\big) > \delta \mid D\big]. \quad (2)$$

For a finite posterior this is the sum of the weights of hypotheses outside the tolerance. It can be large even for an excellent image fit. Its calibration depends on the forward models and sensor noise represented in the posterior.

Let s be a candidate measurement, y_s its possible outcome and c_s its cost in units of threshold failure. We choose the lowest-cost option from

$$J(\text{proceed}) = r(D, a_\star), \quad (3)$$

$$J(s) = c_s + \mathbb{E}_{y_s|D,s}\big[r(D \cup \{y_s\}, a_\star)\big]. \quad (4)$$

For each hypothetical observation we update *both* the weights and the point prediction. The expectation uses possible measurement outcomes available from the model, never the true future response. The beam experiment uses threshold loss and different costs for looking and pushing. The spatial field experiment evaluates an analogous squared-error design for a fixed number of scalar measurements.

4. Physical Explanations and Observation Models

Spatial material, support and load. Figure 2 shows the unknown-load case. In the joint-field study the object is a known, small-strain elastic volume with displacement u_t from its rest configuration. Its positive Young's modulus is $E(X) = \exp(m_0 + m_1\chi(X))$ at material coordinate X, where χ is the volume-centred difference of two normalized Gaussian features. Two additional Gaussian surface footprints describe a compliant support and an applied traction. Each surface footprint integrates to one over its surface. Its position describes *where* the interaction acts, while each traction amplitude gives a total force in newtons. The support has stiffness k_b and energy $\frac{1}{2}k_b \int_{\Gamma_b} w_b(X)\|u\|^2 dA$. Two signed traction modes point downward on the top and laterally on a side. They share an inferred longitudinal centre c_f. The field hypothesis is $\theta = (m_0, m_1, c_b, \log k_b, c_f)$.

For each candidate, the tetrahedral stiffness operator K_θ and the unit-load map B_θ give

$$K_\theta u_t = B_\theta a_t + f_t^{\text{known}}, \qquad y = H_\theta a + d_\theta + \epsilon. \tag{5}$$

Here a stacks two force amplitudes at each of eight times. Solves for unit loads and projection into calibrated cameras define H_θ; d_θ is the known-load contribution. Scaling material stiffness, support stiffness and an unknown force by the same factor preserves the unknown-load displacement. An independently calibrated force breaks this symmetry. A Gaussian support footprint locates a mechanical interaction; a rendering Gaussian carries appearance. Only the former enters the stiffness operator.

Marginalizing unknown load histories. Conditional on θ, let $a \sim \mathcal{N}(0, \Lambda)$ with temporally correlated amplitudes and $\epsilon \sim \mathcal{N}(0, R)$. Gaussian integration gives

$$C_\theta = R + H_\theta \Lambda H_\theta^\top, \tag{6}$$

$$p(y \mid \theta) = \mathcal{N}(y; d_\theta, C_\theta), \tag{7}$$

$$\mu_\theta = \Lambda H_\theta^\top C_\theta^{-1}(y - d_\theta), \tag{8}$$

$$S_\theta = \Lambda - \Lambda H_\theta^\top C_\theta^{-1} H_\theta \Lambda. \tag{9}$$

The last two quantities describe the conditional force posterior. The likelihood includes $\log|C_\theta|$ as well as the residual. Thus a stiff body can fit the same measured displacement as a soft body while requiring a different force distribution. Predicting a hidden displacement under the original load uses the corresponding force conditional; a known reset load requires a forward solve for each material/support hypothesis. In both cases we integrate over the posterior weights. The unknown-force calculation applies where the observation model is affine in force amplitudes. Contact and impact forecasts below use known commands and direct rollouts instead.

An enumerated 5^5 dictionary provides exact weights for its 3,125 spatial hypotheses. An independent implementation uses adaptive tempered sequential Monte Carlo in the five continuous coordinates [4], retaining the Gaussian force conditional. The continuous study replays the finite planner's actions. A hierarchical variant additionally infers one of five force amplitude scales under a Matérn-$3/2$ time prior. This accounts for a force prior itself supplying apparent information about stiffness. Discretization, priors, and sensitivity to temporal mismatch are given in the supplement.

A view and a calibrated reset. For the spatial field study the target concatenates hidden displacements under the original loads and responses to four *new* known loads, with 192 scalar coordinates in each group. The design minimizes expected coordinate variance averaged over this fixed target. A new projected track on the original episode updates both the physical weights and conditional forces. A reset probe supplies a response to a known load and updates the physical weights alone. We allow four scalar measurements and approximate each one-step predictive expectation with 128 common-random-number draws. For the separate beam decision study, bending and base compliance replace the five spatial coordinates. A single tip observation constrains their sum, while either a view of the moving support or a calibrated push at another location can distinguish their contributions.

From video to dynamic response. The dynamic trials take calibrated cameras, a metric rest shape, and known load commands. The earlier RGB-D cantilever study follows surface texture with pyramidal Lucas–Kanade tracking initialized from known rest points; noisy depth supplies the third coordinate. The completed RGB-only benchmark instead follows each point in two overlapping fitting cameras and triangulates its position. Geometry and rest-point initialization are additional inputs to our estimator. GIC and PAC-NeRF reconstruct their object from video. The reserved camera participates only in evaluation.

For the cantilever, each candidate specifies E/ρ, ρ and Poisson's ratio; the support is clamped. A bank of coarse linear elastodynamic responses predicts motion under gravity and an applied force. With observed marker displacements y_i, predicted displacements $g_i(\theta)$ and observation noise σ,

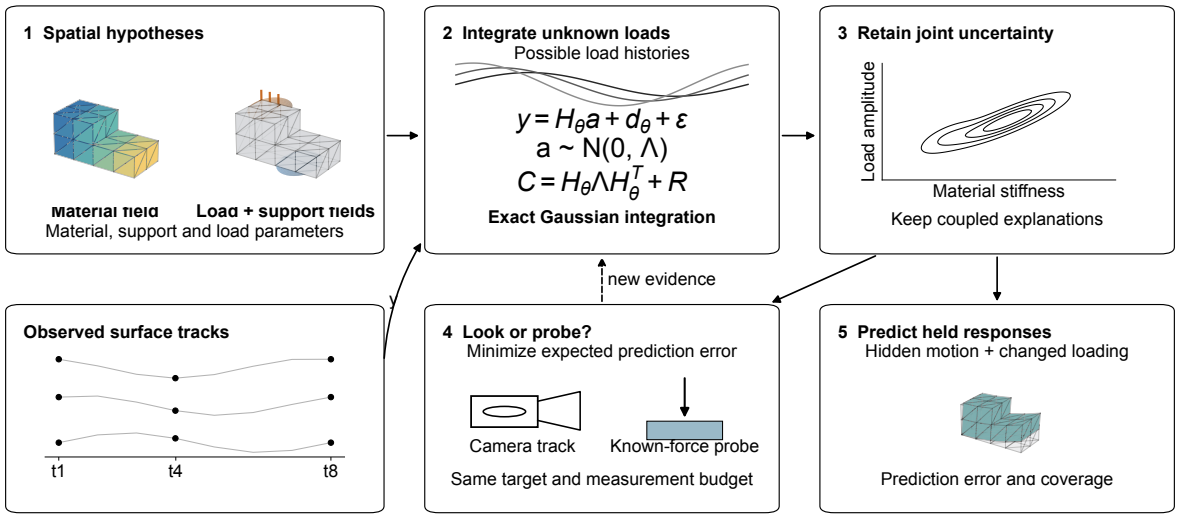


Figure 2. **The unknown-load branch of the method.** Each spatial hypothesis defines a response to unit forces. Integrating over force amplitudes assigns weights to material and support hypotheses and preserves their conditional force distributions. Candidate views and probes are compared in terms of the resulting change in predictive uncertainty. In the known-command dynamic experiments we use the measured command directly.

the weights after each episode satisfy

$$w_\theta \propto p(\theta) \exp\left[-\frac{1}{2\sigma^2} \sum_i \|y_i - g_i(\theta)\|^2\right]. \qquad (10)$$

We keep the noise estimate from the initial fit for the calibration update. We apply the held-out command to each weighted hypothesis and compute its predictive mean and threshold risk. The RGB-only block protocol uses an analogous bank over inclusion position, stiffness contrast, background modulus and Poisson ratio, with a homogeneous block also permitted. The compared methods have different physical model families; full adapter contracts and numerical controls are in the supplementary. The striker protocol propagates the calibrated cantilever posterior through a compliant contact law and scores momentum transfer from the freely evolving sphere. Each comparator receives the same prescribed calibration episode in these three scenarios; selective acquisition is evaluated separately.

5. Experiments and Results

5.1. Three prospective interactions predicted from RGB

Our benchmark asks whether a model fitted to an observed motion episode D_0 and optionally updated with one calibrated intervention D_1 can predict an independent, held-out action $a_\star$. Ten calibrated cameras provide fitting RGB; camera 0 is reserved for evaluation. Our estimator triangulates RGB tracks from two cameras, uses a known rest geometry, updates a posterior over mechanical hypotheses, and forwards each hypothesis under the test action. Baseline methods fit or adapt their own mechanical representations using the same evidence episodes. The held-out reference remains unavailable until the predictions are sealed. The supplement provides the adapter contracts, provenance and complete case scores.

Three tasks probe complementary predictions. **S1: Deformable cantilever, changed force direction.** Three cantilevers have equal gravity-driven motion because mass density and elasticity scale proportionally. The test applies a new 20 N diagonal force at $0.75L$, and we score 3D station displacement over 37 frames. **S2: Elastic block with hidden stiffness.** A visually homogeneous block conceals an eightfold-stiffer inclusion; a new 30 N off-centre press is evaluated at 27 surface positions over 25 frames. **S3: Free striker collision.** The cantilever family is struck by a freely rebounding 0.30 kg sphere at 1 m/s; the primary score is absolute axial impulse error, with cantilever displacement as a complementary response metric. There are three reference worlds per task. S1 and S3 share the same three physical objects but test different interactions.

Changed-direction cantilever push. From D_0 alone, our future-action station error is 6.94 mm, compared with 5.87 mm for the strongest baseline. The known-force calibration reduces our error to 1.27 mm (2.18, 1.01 and 0.63 mm in the three worlds). After the same calibration, M-PhyGs,

Table 1. **Prediction of new physical interactions after one calibrated observation.** Mean over three synthetic material/inclusion worlds per task (lower is better). All figures use the RGB-only evaluation. The S1, S2, and S3 displacement columns report 3D RMS error in mm; the final column is absolute transferred-impulse error in N s.

Method	S1: cantilever RMSE (mm)	S2: block RMSE (mm)	S3: cantilever RMSE (mm)	S3: impulse Error (N s)
Ours (RGB)	**1.27**	**0.33**	**0.51**	**0.003**
M-PhyGs [23]	5.26	1.28	3.47	0.073
PhysTwin [9]	5.46	1.73	5.92	0.086
GIC [3]	7.21	2.60	7.97	0.069
PAC-NeRF [15]	9.45	1.78	8.27	1.771

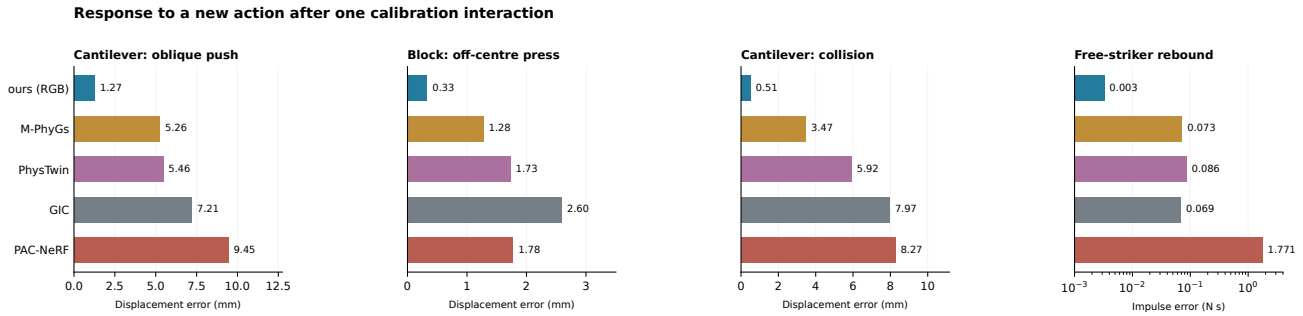


Figure 3. **Held-out action response error.** Mean errors for the five compared methods, after one known calibration action. The collision impulse panel uses a logarithmic scale. Each bar combines three worlds; per-world values and the effects of the evidence update are tabulated in the supplement.

PhysTwin, GIC and PAC-NeRF score 5.26, 5.46, 7.21 and 9.45 mm. The target direction differs from both passive gravity and calibration, so the improvement measures transfer to a new force, not re-rendering of the observed action.

Unseen loading of a heterogeneous block. Our spatial material-hypothesis bank attains 0.33 mm mean error, compared with 1.28 mm for the next comparator. The initial central press already concentrates our finite posterior on one hypothesis; the additional calibration press makes no difference to the resulting prediction. This task therefore probes evidence sufficiency and a spatially resolved physical explanation, rather than an incremental improvement from the second measurement.

Striker momentum and deformation. For the free-impact test, calibration reduces our cantilever station error from 2.40 to 0.51 mm and our absolute striker impulse error from 0.042 to 0.003 N s. The smallest competing post-calibration station error is 3.47 mm (M-PhyGs), and the smallest competing impulse error is 0.069 N s (GIC). The impulse endpoint tests the mechanical consequences of impact rather than only the visible shape trajectory.

Held-out camera rendering. We render all sealed motion forecasts from the reserved camera using one shared GIC-learned Gaussian appearance. This controls the appearance field when measuring the effect of motion. A second, reference-texture visualization transfers the same predictions to the known mesh to increase sensitivity to deformation error. In the shared-appearance test, our post-calibration score is 21.62 dB on S1 versus 21.64 dB for reference motion, and 21.75 dB on S3 versus 21.70 dB for reference motion. On the block all predictions remain near the shared-appearance 20.72 dB ceiling even when their displacement errors differ considerably. The reference-texture block rendering instead scores 34.65 dB for our predicted motion against 28.16 dB for the best other post-calibration motion. This distinction directly demonstrates that rendering fidelity alone can obscure differences in the underlying physical response.

5.2. Selecting whether to proceed, look, or probe

We independently evaluate evidence acquisition with a compliant-base cantilever model. Ninety-six physical worlds each face three target/visibility settings. Given the initial noisy tip observation, the policy proceeds without measurement in all 96 repeated-load cases (0.104 mm RMSE). If the target load changes, it requests a view in 92 of 96 cases when base motion is observable (0.073 mm final RMSE),

or a calibrated push in 91 of 96 cases when the base is hidden (0.119 mm final RMSE). View and probe costs are 0.015 and 0.040 in threshold-failure units at a 0.50 mm tolerance. The mean decision-loss advantage over a cost-aware Gaussian-moment policy is small, 0.00033, with a bootstrap confidence interval ending at zero. These are fixed-simulator decision experiments; the RGB benchmark above instead applies the same prescribed calibration to every comparator.

5.3. Additional independent physical controls

The original torus continuation experiment separates a time-conditioned appearance forecast from a physics rollout: the GIC deformation network reaches 20.47 dB on observed motion but 8.29 dB on future frames, whereas GIC physics and PAC-NeRF achieve 18.44 and 18.01 dB on that continuation. In a separate three-object drop control, scaling mass, stiffness and damping together yields almost indistinguishable passive movies (94.6–100.1 dB between simulated movies), but different known-force responses. A contact-model stress experiment estimates 1.86% risk of exceeding a 3 mm tolerance while realizing 43.75% failures when tested against a finer reference. These controls motivate both task-conditioned forecasts and explicit treatment of model mismatch.

5.4. Joint spatial identification under unknown loading

We also evaluate when a view or a known force provides better evidence in the presence of unknown loading. We use a block and stepped block of bounding dimensions $120 \times 60 \times 60$ mm. Inference starts from 24 scalar observations of projected motion on the visible surface: four material points, two orthographic components and three loading times. Additional tracks use an oblique camera; a geometric visibility check covers all deformed test states. Sensor noise is 0.02 mm. The force standard deviation and calibrated probe amplitude are 0.08 N. The two geometry-specific development sets select fixed schedules before held-out evaluation. The supplement describes the discretization and all independent cohorts.

Acquisition and the target. Across 128 matched cases, four adaptive measurements reduce combined coordinate RMSE from 0.0578 to 0.0340 mm. Fixed camera-only, fixed mixed and random policies reach 0.0403, 0.0423 and 0.0438 mm. Against the fixed mixed schedule, the paired difference is -0.0083 mm (95% case-bootstrap interval $[-0.0110, -0.0058]$). The adaptive policy spends 439 of its 512 measurements on camera observations. Probes alone give the best prediction under known new loads, at 0.0099 mm against 0.0144 mm for adaptive selection, but leave more uncertainty in the original hidden motion (Table 2, Fig. 5). In another 128 cases with uncertain force scale, four fixed views reduce hidden-motion error by 33.4%, while four calibrated probes reduce new-load error by 48.4%. Both branches start with the same evidence.

Table 2. **Prediction after four scalar observations.** Means over 128 matched cases. Hidden and new-load errors are coordinate RMSE in mm. Material error is relative; force error is in newtons. Fixed denotes the development-selected mixed schedule.

Policy	Hidden mm	New load mm	Material rel.	Force N
Initial	0.076	0.024	0.451	0.042
Active	0.045	0.014	0.347	0.030
Fixed	0.057	0.012	0.322	0.033
Fixed View	0.050	0.022	0.421	0.032
Fixed Probe	0.074	0.010	0.277	0.040
Random	0.059	0.014	0.343	0.034

What retaining dependence changes. Retaining the conditional force distribution for each material/support hypothesis gives a mean prediction-interval width of 0.0717 mm; replacing those conditionals with a common moment-matched distribution gives 0.0863 mm. Coverage is 92.0% versus 93.9% at nominal 90%, and hidden-motion errors are nearly equal (0.0446 and 0.0449 mm). In this experiment the practical gain is narrower intervals at similar accuracy. On a separate continuous-prior cohort, sequential Monte Carlo gives 0.0294 mm after the recorded adaptive actions versus 0.0350 mm after fixed mixed observations.

Calibration outside the fitted family. Finite-model prediction intervals cover 92.0% of matched target coordinates. Coverage falls to 86.4% for continuous parameters, 86.1% when the generator uses a finer mesh, and 79.2% for a shorter correlation length in the true force history. A follow-up that mixed the temporal length and amplitude on 64 fresh stress cases improved force coverage from 72.4% to 83.9%; inferred material error increased. More observations can narrow a posterior around the wrong explanation.

6. Discussion

A visually persuasive reconstruction does not imply that the response to an unobserved action is identified. The relevant uncertainty is the disagreement among mechanically plausible worlds *under the intended interaction*. A passive video can be insensitive to material scale; a known force can resolve that ambiguity. Conversely, a repeated predictable action need not justify another measurement. The calibrated response and the choice of whether to measure are therefore different aspects of the same task-conditioned inference problem.

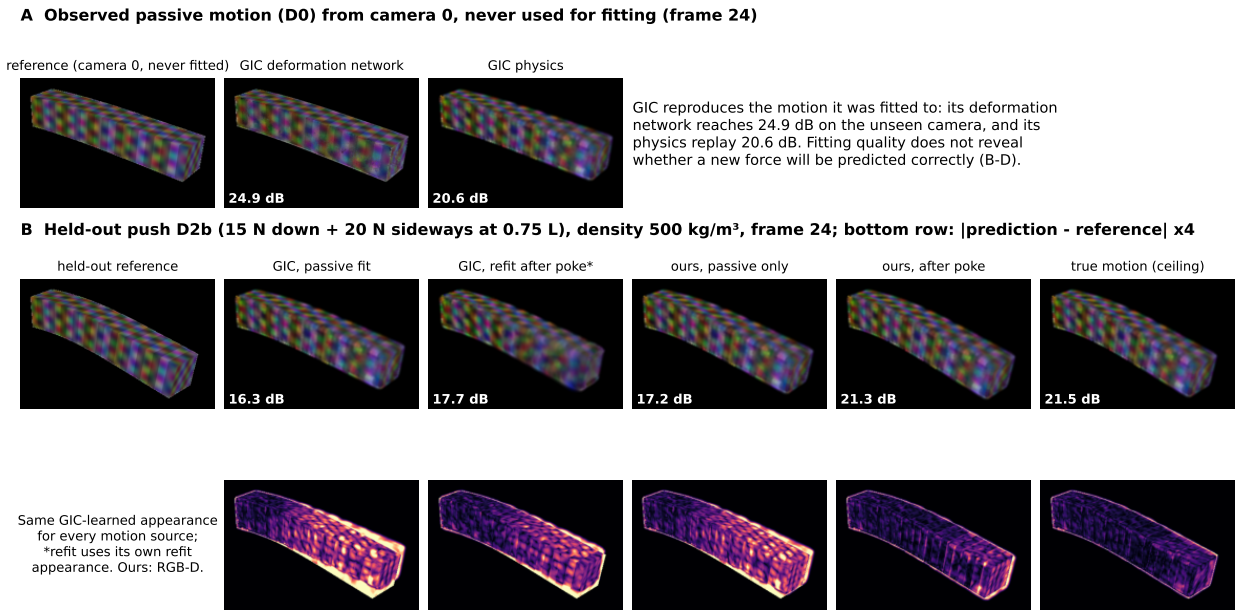


Figure 4. **Rendering the measured and predicted cantilever motion from a withheld camera.** The upper row shows a passive frame. The lower panels show the new diagonal push on the light cantilever with pixel differences magnified fourfold. Numbers in the images are sequence foreground PSNR. Comparisons primarily use one frozen GIC appearance; the GIC refit uses its own appearance. The motion predictions were sealed before reference generation; the common-appearance analysis was finalized subsequently.

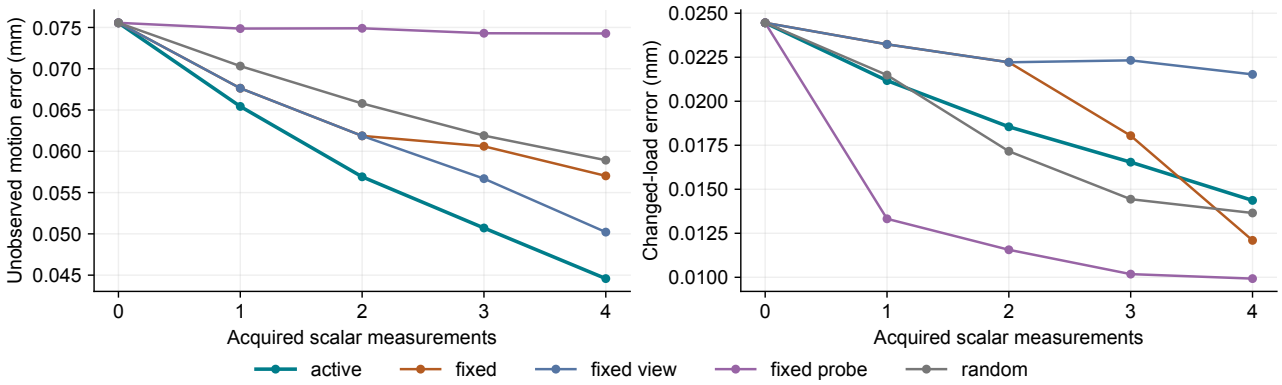


Figure 5. **Measurement choice depends on the target.** An additional camera can expose the original hidden motion, whereas a known probe directly constrains response under a changed load. Curves average all matched surface-track cases.

The prospective RGB benchmark assesses predictions under a changed force direction, a hidden spatial stiffness pattern and an unconstrained contact outcome. Its direct measurements of 3D deformation and transferred momentum complement held-out-camera rendering. The latter can show excellent image fidelity even when motion errors differ, especially when errors in appearance dominate small deformations. The independent evidence-selection experiment evaluates the choice to proceed, look or probe rather than requiring the same calibration in every case.

The supplementary material documents the benchmark adapters, matched-model assumptions, input provenance, sensitivity analyses, computational costs and numerical controls. These define the present controlled results and the validation necessary for physically distinct or real-world objects.

7. Conclusion

Physical world models should be evaluated by their predicted response to the next interaction, not solely by the accuracy of reconstructed video. Our action-conditioned representation of mechanical ambiguity yields testable deformation and

impulse forecasts, and it motivates collecting a view or a probe only when that evidence matters. The completed RGB intervention benchmark and independent measurement studies establish these complementary claims in controlled synthetic settings.

LLM Usage Statement

OpenAI ChatGPT and coding agents, and Anthropic Claude, assisted with implementation, synthetic experiment orchestration, numerical and plotting scripts, literature organization, and manuscript preparation. ChatGPT was also used for reviewer-style feedback, exploration of experimental hypotheses, and editing and drafting text. Numerical figures were produced by scripts from simulation outputs or recorded scores. The authors take responsibility for the submitted text, references, code, and results.

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